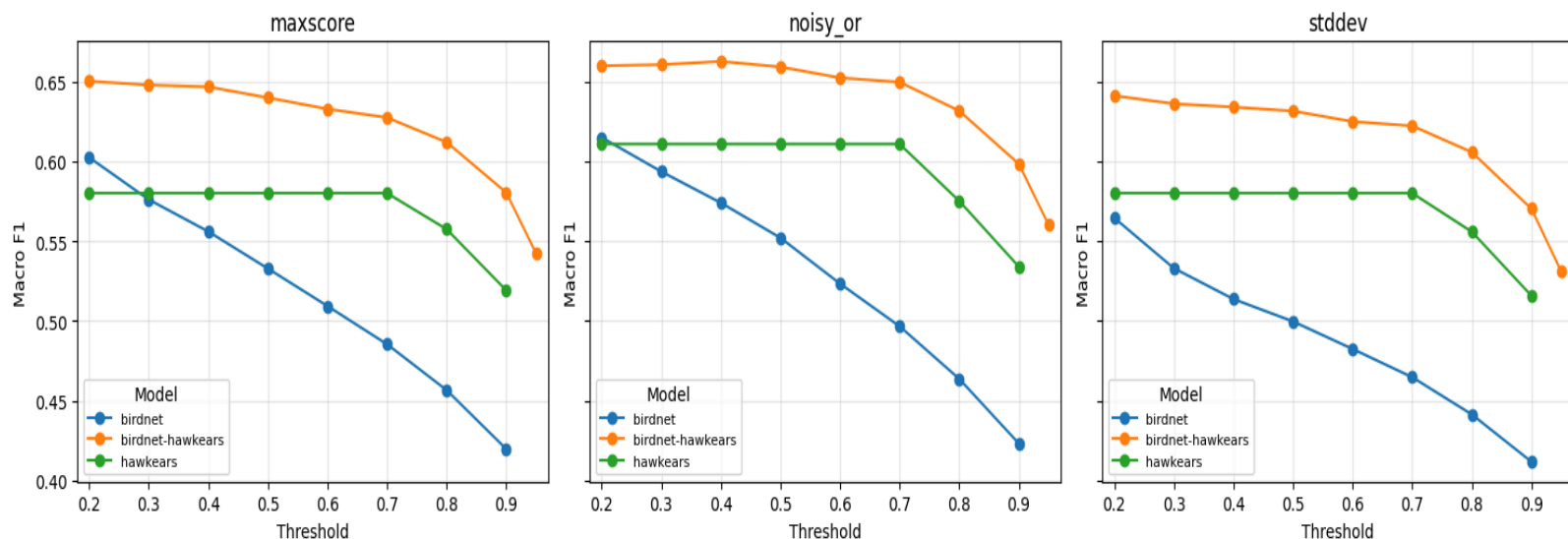


Ecdysis Foundation is a non-profit focused on regenerative agriculture. Biodiversity is a valuable measure of regenerative ecosystem health, and Ecdysis wants to use machine learning to automate bird identification using over 1000 bird audio recordings.

We improved bird detection using a multi-stage pipeline and two audio classification models (HawkEars and BirdNET). First, we filtered out audio too low or too high to be bird calls and reduced background noise. Next, the models identified which species were present in the filtered audio, when, and their certainty probabilities.

Because probabilities varied greatly by species, we converted bird species-specific scores into simple “bird” vs “no-bird” probability scores with different methods (i.e., max score, Noisy-OR, or standard deviation). We evaluated which methods’ outputs best matched expert observations using F1. The visualization below shows the F1 scores across different probability thresholds for each method and model. Threshold is the cutoff point for when a prediction score indicates “bird” vs “no-bird”.

Macro F1 vs Threshold by Postprocessor



The combination of frequency filtering, noise reduction, running both models, and applying Noisy-OR with threshold 0.4 was most effective against ground-truth data. Moving forward, Ecdysis can implement this pipeline to reduce the amount of audio experts need to parse to evaluate biodiversity.